Alessandro Pani Patiño

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Summary

Computer Science student at the University of Costa Rica who builds and ships production software: a web app that replaced a paper process for 50+ UCR staff, and websites for 10+ freelance clients since 2022. Bilingual (Spanish/English); works in multidisciplinary Agile (Scrum) teams. Available full-time from December 2026.

Education

University of Costa Rica | B.S. Computer Science (Software Eng.) *Expected completion Dec 2027*

Experience

Research & Development Assistant | Ensamblaje Piñero, UCR *Jun 2026 - Present*

- Sole developer on a UCR linguistics research project: built Python pipelines that collected and segmented 7,063 Legislative Assembly transcripts (~17 GB) into a traceable corpus, plus user guides for the team.
- Building a benchmark of Spanish part-of-speech taggers, from pretrained pipelines (Stanza, spaCy) to a fine-tuned transformer language model (BETO), with a strict EAGLES evaluator and 67 automated tests.

Full-Stack Developer (paid student assistantship) | SIGECAT, UCR *Jan 2026 - Jun 2026*

- Replaced paper-based work-schedule and shift declarations with a production web application used by 50+ university staff; built and integrated the React + TypeScript frontend, PHP REST API and Oracle data layer.
- Cut slow Oracle query times from about 4 s to under 1 s; verified the API with endpoint and concurrency tests.
- Secured the API (token rotation, bcrypt, rate limiting, lockout); shipped with Docker and CI on Oracle Cloud.

Freelance Web Developer *2022 - Present*

- Built websites for 10+ small businesses (legal, hospitality, e-commerce) that give them a professional image and help them win new customers, covering requirements, deployment and SEO.
- Troubleshoot client issues, configure email, DNS, SSL and hosting, and train clients to edit their content.

Software Developer (community service, TCU) | deferia.cr, UCR and FAO *Aug 2023 - May 2025*

- Joint UCR and UN FAO project: turned four years of CNP price bulletins into a clean 9,184-record dataset across 56 products (Django, PostgreSQL) and built interactive price charts (React, Plotly).

Projects

Mods | AR Custom Cap Store, UCR Software Quality course *mods-cr.vercel.app*

- Built a web store where customers design a trucker cap and try it on in 3D with their phone camera; face tracking runs on-device, so no image leaves the phone (TypeScript, Three.js, MediaPipe).
- Tested with 146 unit tests (Vitest) and Playwright end-to-end tests (privacy, offline, checkout); designed a GQM plan of 15 code and process quality metrics and 16 V&V activities.

ECCI QR Access Control, UCR *ecci-qr-access.vercel.app*

- Built a QR check-out system to replace the CS school's paper device log, designed to shorten peak-hour exit lines; every entry and exit is timestamped per device and user (FastAPI, PostgreSQL, React).

Museo del Espacio Matemático | 30+ WebGL simulations, hand tracking *Live demo*

Skills

Languages: Python, TypeScript/JavaScript, PHP, SQL (Oracle, PostgreSQL, SQLite)
Web: React, Django, FastAPI, Three.js, REST APIs
Data and ML: pandas, scikit-learn, Power BI, PyTorch, Hugging Face Transformers, spaCy, Stanza
Tools: Git, Docker, Linux/Bash, pytest, Vitest, Playwright, CI pipelines, Jira, Agile/Scrum, Oracle Cloud
Spoken: Spanish and Italian (native), English (C1, professional)